Alpha Research Final Project

Factor-Neutral Long/Short Equity Strategy

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Research Pipeline

Six sequential steps — each output feeds directly into the next.

1	2	3	4	5	6
Data	**PIT Treatment**	**Factor Neutralisation**	**Signal Construction**	**Portfolio Construction**	**OOS Test**
Bloomberg & yfinance	Observation-date anchoring	FF3 + sector demean	Build & validate	Equal-wt → Inv-vol → MVO	2023–2026 period

Objective : *A market-neutral equity strategy pursuing alpha diversification across signal families, neutrality to common risk factors, regime-robust feature selection via rolling ElasticNet, and efficient risk-adjusted weight allocation via MVO*

Data Hygiene & Bias Control

Backtest credibility depends on rigorous data engineering before any signal is built.

✓	**Point-in-time anchoring**	Every field enters only on its public observation date
✓	**Staleness limits**	Fields expire past their refresh cycle — 14 to 120 days
✓	**Tradability screen**	Zero volume, non-positive price, stale-price excluded
✓	**Corporate-action cleaning**	Recycled tickers and wrong-security histories removed
✓	**Survivorship-free universe**	S&P 500 membership reconstructed month-by-month

Factor Neutralisation

Residualise returns against common factor and sector exposures; all subsequent signal work uses these residuals, not raw returns.

Step 1 — Rolling Fama-French 3-factor regression

- Two-year rolling regression of winsorized returns against market, size, and value factors
- Betas estimated on prior data only ensuring no look-ahead
- Residuals are the factor-stripped return used as signal targets
- Stocks with under two years of history excluded

Step 2 — GICS sector demean

- Residual demeaned within each GICS sector
- Removes residual sector tilt not captured by the FF3 factors

Signal Library I

Eleven candidate signals across five families and two interactions.

MOMENTUM

12-Month Momentum

Cumulative log-return t−52w to t−4w (skip last month).

EARNINGS

Post-Earnings Drift

Standardised earnings surprise at announcement.

Estimate Revisions

4-week percentage change in consensus EPS estimate.

QUALITY

Gross Profitability

Gross profit / total assets (Bloomberg).

Accruals Quality

− (net income − cash flow from operations) / assets. Lower accruals = higher earnings quality (Bloomberg).

Return on Equity

Net income / common equity (Bloomberg RETURN_COM_EQY field).

Signals are winsorised (1st–99th percentile) and z-scored cross-sectionally per date before compositing.

Signal Library II

Continued — value, short-interest, and interaction families.

VALUE

Earnings Yield

Trailing twelve-month EPS / price (Bloomberg). Recomputed daily as price moves.

SHORT INTEREST

Short Interest Level

Shares short / shares outstanding (Bloomberg Short Interest Level field).

Short Interest Change

4-week change in short interest level.

INTERACTIONS

Momentum × Short Interest Change

Only positive momentum + falling short interest AND negative momentum + rising short interest preserved. Product of cross-sectional z-scores.

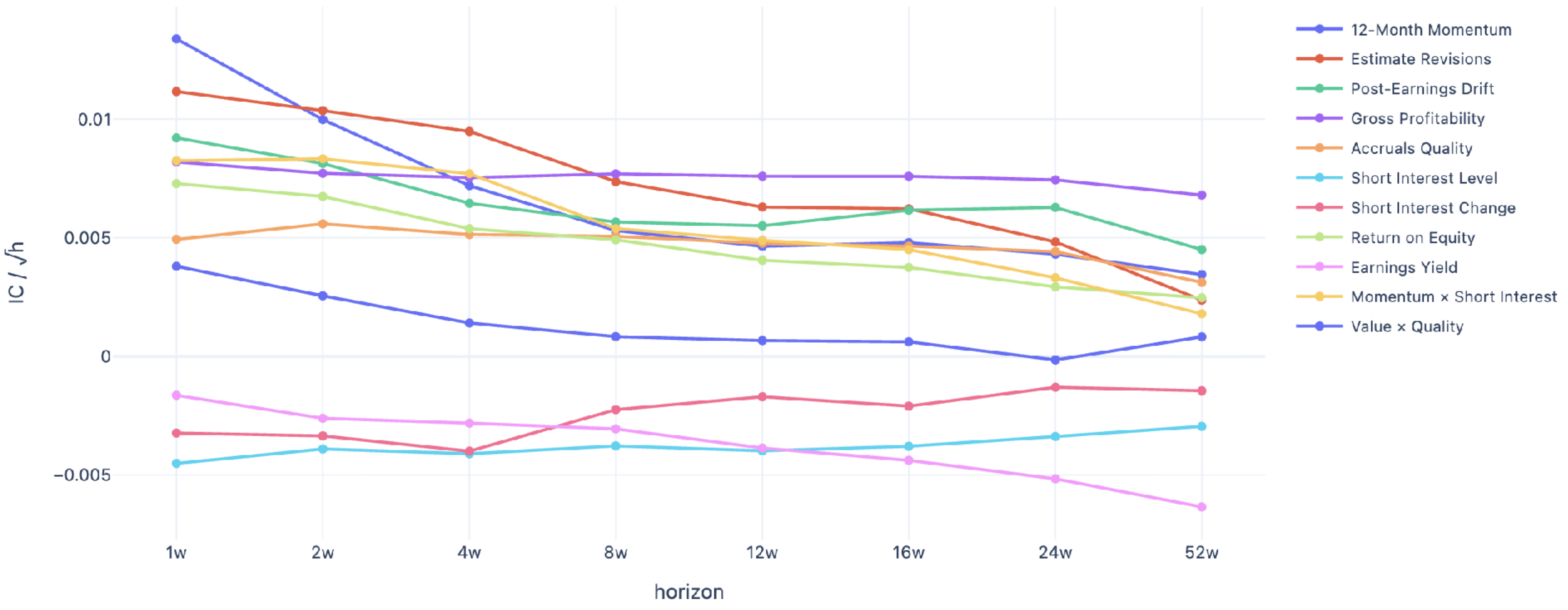
Value × Quality

Earnings Yield × Return on Equity (Bloomberg). Product of z-scores; long cheap, high-quality. Bullish leg preserved — cheap (positive earnings-yield z-score) AND high quality (positive ROE z-score).

Signals are winsorised (1st–99th percentile) and z-scored cross-sectionally per date before compositing.

Signal Evidence — Predictive Horizon

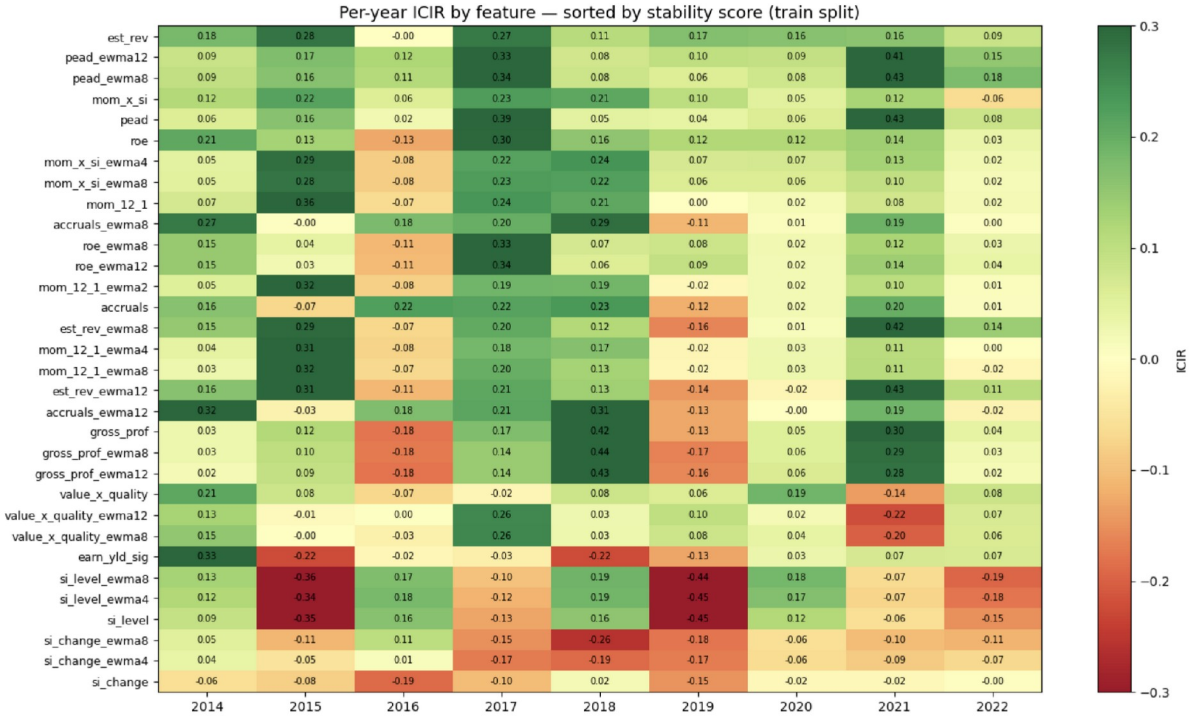
Predictive power is strongest at short horizons, supporting weekly rebalancing.



IC / √h is the information ratio per unit of time. A flat or declining curve means the edge does not improve with longer holds.

Signal Stability

ICIR measures the stability of each signal's cross-sectional predictive power over time.



What the heatmap shows

- Each cell reports the annual ICIR for one signal in one calendar year.
- ICIR is the ratio of mean weekly Information Coefficient (IC) to the standard deviation of weekly IC within that year.

Diagnostic only: the heatmap describes signal stability; it does not assume that every signal is consistently predictive every year.

Data Timeline & Sample Split

The sample is separated into estimator burn-in, model selection, and final out-of-sample evaluation through April 2026.

2012–2014

Burn-in: factor estimation

Two-year rolling Fama-French regressions initialise stock-level factor exposures and residual return targets.

2014–2016

Burn-in: ElasticNet initialisation

The feature history accumulates until the first 3-year rolling ElasticNet window is available.

2017–2022

Training and model selection

Train data is used for performance analysis and MVO hyperparameter grid search.

2023–Apr 2026

Out-of-sample test

Final evaluation is reported on untouched OOS data through April 2026, net of transaction costs.

Burn-in periods are consumed to initialise rolling estimators and are excluded from reported performance. Train results guide model selection; OOS results judge final performance through April 2026.

Signal Construction — Rolling ElasticNet

Signal features are engineered first; ElasticNet then learns feature coefficients using only past data.

Raw signals
11 candidate signals

›

Feature expansion
EWMA variants
2, 4, 8, 12 weeks

›

Feature set
32 cross-sectional features

›

Rolling ElasticNet
Monthly refit on prior 3 years

›

Cross-sectional prediction
Ranked and standardised by date

Rolling estimation window

At each monthly refit, the model uses the prior 3 years of pooled cross-sectional observations.

Train: 2.5 years	Validation: 0.5 years

Validation period uses the final 0.5 year of the rolling window.

Hyperparameter and coefficients

The validation split selects the regularisation strength. The model is then refit, producing feature coefficients and a standardised cross-sectional prediction.

ElasticNet Deployment IC

Rolling out-of-sample Information Coefficient by monthly refit, 2017–2022.

Read

— Mean IC is positive (0.0139), indicating predictive content on average.

— IC varies across refit months, so the signal is informative but not uniform through time.

Positive but noisy IC across deployment windows; used as a rolling validation diagnostic.

Portfolio Construction

Three levels of increasing sophistication — all share the same neutrality constraints.

1 Equal Weight

Top/bottom quintile selection; equal gross allocation.

Quintile cutoff	Annual return	Net Sharpe	Avg turnover /reb.	Annual turnover
20%	2.3%	0.36	43.5%	2261%
30%	1.5%	0.30	39.3%	2045%
50%	0.8%	0.23	32.4%	1685%

→

2 Inverse Volatility

Same long/short selection; weights are inverse to residual volatility.

Quintile cutoff	Annual return	Net Sharpe	Avg turnover /reb.	Annual turnover
20%	1.7%	0.30	45.5%	2367%
30%	0.9%	0.20	40.7%	2118%
50%	0.1%	0.04	32.8%	1706%

→

3 Mean-Variance Optimization

Grid-tested parameters

- λ (lambda): 1.0, 5.0
- γ (gamma): 0.0, 1.0, 3.0
- Per-stock cap: 0.5%, 1%, 2%

Results for the grid search are reported on the next slide.

Default transaction cost assumption: 5 bps.

MVO Grid Search — Full Results

Sorted by train Sharpe (2017–2022). Selected specification highlighted: λ=5, cap=0.5%, γ=1.0 — Sharpe 0.82 train

λ	cap	γ	Sharpe	Ann. return	Ann. vol	Max DD	Avg tov/reb.	Tov ann.
5	0.5%	3.0	1.01	3.5%	3.4%	−3.4%	1.4%	73%
1	0.5%	3.0	1.00	3.4%	3.4%	−3.4%	1.4%	73%
1	1.0%	3.0	0.85	4.6%	5.4%	−5.3%	2.2%	114%
5	1.0%	3.0	0.84	4.5%	5.3%	−5.2%	2.2%	114%
5	**0.5%**	**1.0**	**0.82**	**4.0%**	**4.9%**	**−5.6%**	**6.0%**	**313%**
1	0.5%	1.0	0.80	3.9%	4.9%	−5.3%	6.0%	312%
5	2.0%	3.0	0.77	5.9%	7.6%	−7.2%	3.3%	174%
5	1.0%	1.0	0.73	5.5%	7.5%	−7.5%	9.4%	490%
1	1.0%	1.0	0.73	5.5%	7.6%	−7.5%	9.4%	490%
1	2.0%	3.0	0.70	5.4%	7.7%	−7.9%	3.3%	174%
1	1.0%	0.0	0.54	4.3%	8.0%	−14.5%	46.5%	2419%
5	1.0%	0.0	0.53	4.3%	8.0%	−14.6%	46.5%	2419%
1	2.0%	1.0	0.53	5.4%	10.1%	−11.3%	13.7%	714%
5	2.0%	1.0	0.53	5.4%	10.1%	−10.9%	13.7%	715%
5	0.5%	0.0	0.43	2.2%	5.2%	−11.9%	37.6%	1954%
1	0.5%	0.0	0.42	2.2%	5.2%	−12.1%	37.7%	1960%
5	2.0%	0.0	0.37	4.2%	11.2%	−20.6%	53.4%	2774%
1	2.0%	0.0	0.37	4.2%	11.2%	−20.7%	53.4%	2776%

Annual Sharpe Profile — Train

Annual Sharpe by Year — net of 5 bps transaction cost

Out-of-Sample Results

2023–2026 remains positive net of costs, using 5 bps transaction cost model

1.12

Net Sharpe (test)

7.1%

Annual return

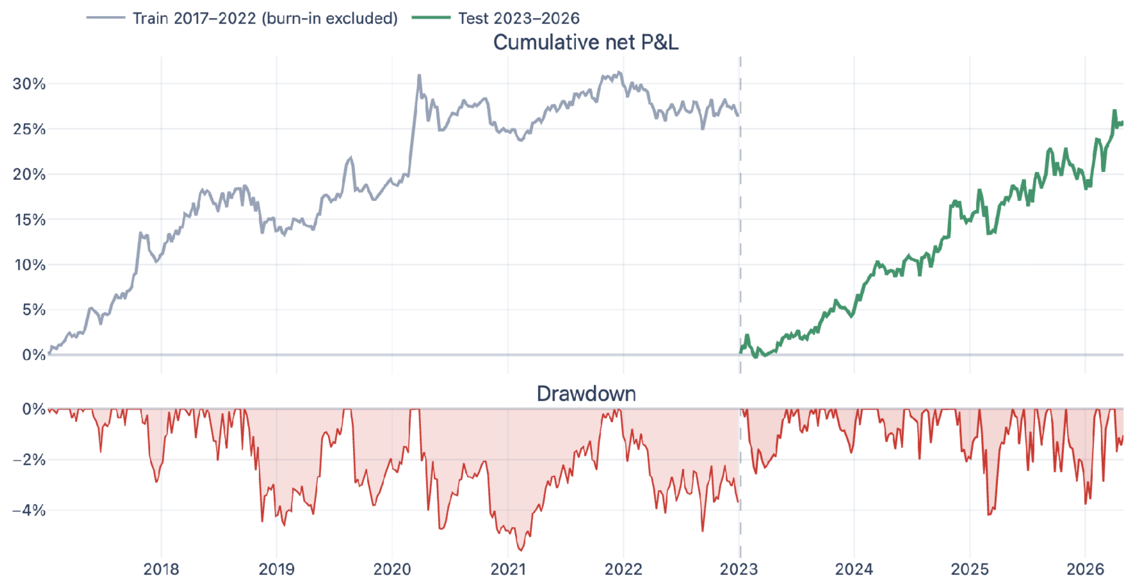
6.3%

Volatility

−4.2%

Max drawdown

Cumulative net P&L and drawdown — frozen model



Average turnover 3.8% per rebalance · ~197% annualised

Annual Sharpe Profile

Net-of-cost Sharpe by year

Read

— 5 of 6 train years positive

— 4 of 4 test years positive

— Performance not concentrated in one year

Factor Exposure — Train vs Test

No FF3 factor loading significant at 5% in either period. Alpha significant OOS ($t = 2.57$).

0.68%

OOS R^2

FF3 factors explain 0.68% of OOS P&L variance

7.29%

OOS alpha

Alpha significant at 5% level (t-stat: 2.57)

none

Significant factor betas

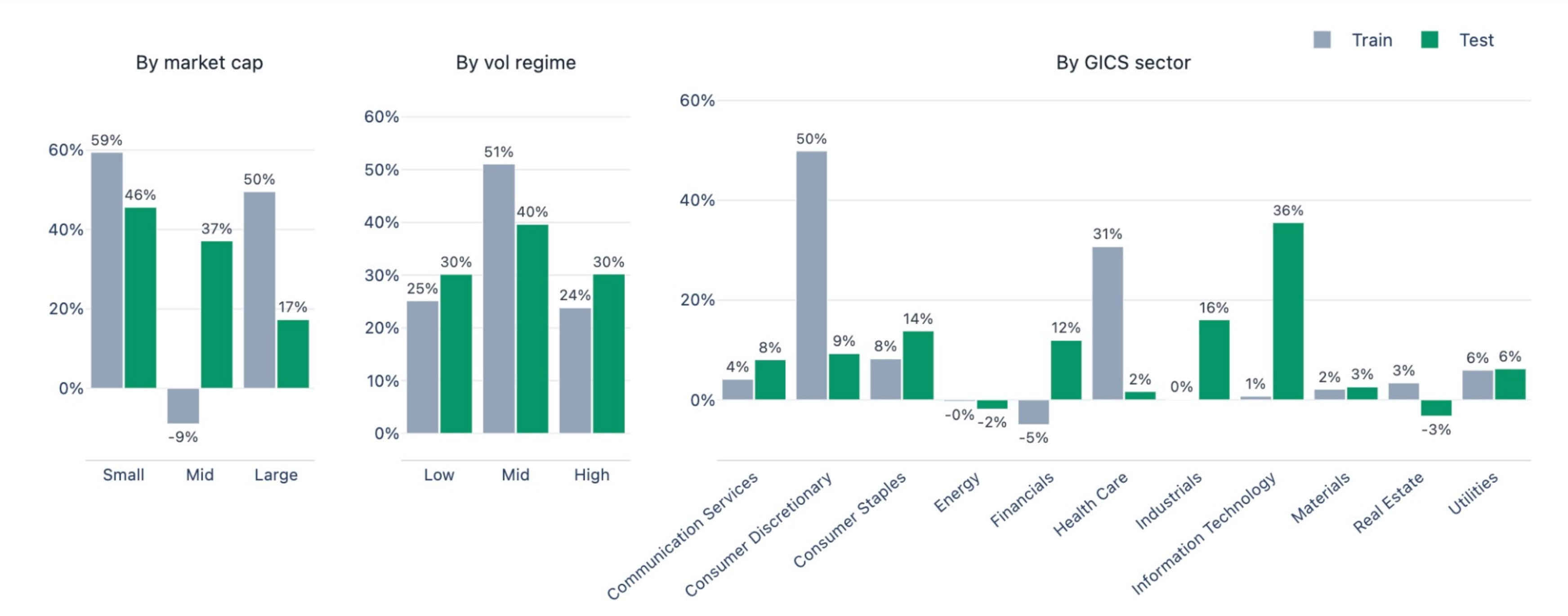
No FF3 loading significant at 5% in either period

Period	Market β	SMB β	HML β	R^2
Train	−0.030	0.026	−0.011	1.58%
Test / OOS	−0.011	−0.038	−0.017	0.68%

- FF3 factors explain 0.68% of OOS P&L variance (train: 1.58%).
- All FF3 betas have $|t| < 1.6$ in both periods — no significant factor tilt.
- Annualised alpha: 7.29% OOS ($t = 2.57$, $p < 0.05$).

Train/Test PnL Attribution Diagnostics

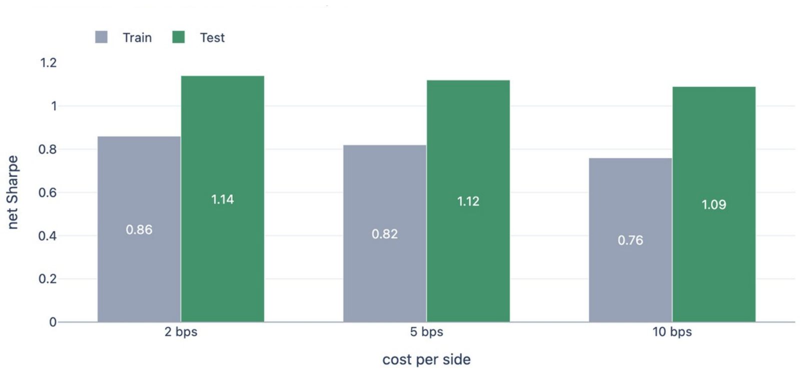
Share of strategy PnL by market cap, volatility regime, and GICS sector — compared across train and test periods.



Robustness — Costs & Capacity

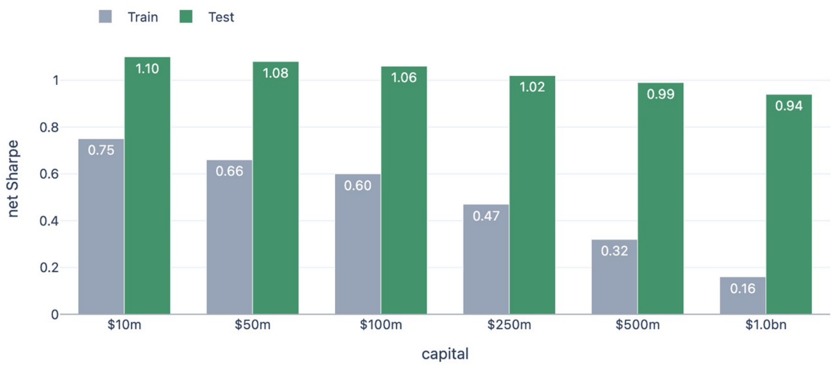
Performance remains positive under conservative implementation assumptions.

Transaction-cost stress



Sharpe 1.00 at 10 bps per side (test).

Capacity stress



Viable to $1bn AUM; Sharpe above 0.90 at $500m.

Market impact uses lagged volume and residual volatility — no future liquidity information enters the test.

Strategy Thesis

A weekly factor-neutral long/short strategy from validated cross-sectional alpha signals.

11 yrs	4 yrs	1.12
Data · 2012–2022	Test · 2023–2026	Net OOS Sharpe

Systematic alpha

11 validated signals across quality, momentum, earnings, short interest

Principled research process

Point-in-time data, factor neutralisation, walk-forward selection

OOS validated

Sharpe 1.12 out-of-sample, net Sharpe above 1.0 at 10 bps and 1bn AUM

Thank you

Questions & Discussion

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Appendix — Key Formulas I

Mean-Variance Optimization and Factor Neutralisation.

Mean-Variance Optimization	$\max_w \; \alpha^\top w \; - \; \lambda \, w^\top \Sigma w \; - \; \kappa \, \mathrm{Tov}_t$	*Maximizes alpha subject to risk and neutrality constraints. Kappa = turnover penalty; Tov = drift-adjusted half-L1 trade size*
Factor Neutralisation (Fama-French 3-factor)	$r_{i,t} - r_f = \alpha_i + \beta_i^{\mathrm{MKT}} \, f_t^{\mathrm{MKT}} + \beta_i^{\mathrm{SMB}} \, f_t^{\mathrm{SMB}} + \beta_i^{\mathrm{HML}} \, f_t^{\mathrm{HML}} + \varepsilon_{i,t}$	*Residual target = return minus fitted factor component.*

Appendix — Key Formulas II

Market Impact (Capacity model)

$$\mathrm{Impact}_{i,t} = |\Delta w_{i,t}|\ \sigma_{i,t}^{20d}\ \sqrt{\frac{|\Delta w_{i,t}|\,\mathrm{NAV}}{\mathrm{ADV}_{i,t}^{20d}}}$$

Uses lagged volume and residual volatility.

Sharpe Ratio (annualised)

$$\mathrm{SR} = \frac{\bar{r}\,\sqrt{52}}{\sigma_r}$$

r-bar = mean weekly net return; sigma = std dev of weekly net return

Appendix — Key Formulas III

Turnover (drift-adjusted)	$\mathrm{Tov}_t = \frac{1}{2}\left\|w_t - \tilde{w}_{t-1}\right\|_1, \qquad \tilde{w}_{i,t-1} = w_{i,t-1}(1 + r_{i,t})$	*w-tilde: prev. weights marked to market (×1+r) before computing trade size*
Net P&L	$\mathrm{PnL}_t = w_t^\top r_{t+1} \; - \; 2 \cdot \mathrm{Tov}_t \cdot c$	*c = 5 bps per side; factor of 2 covers both buy and sell leg*

Appendix — Data Sources

Bloomberg supplies fundamentals and market data; yfinance provides active-ticker prices.

Bloomberg	**Prices & volumes**	88 delisted / renamed tickers for historical coverage
Bloomberg	**Fundamentals**	Net income · Cash from operations · Total assets · Gross profit · ROE · Leverage
Bloomberg	**Valuation**	Earnings yield · Price-to-book, recomputed daily from price
Bloomberg	**Estimates**	Consensus EPS, current and next fiscal year
Bloomberg	**Short interest**	Percentage of float · Short interest ratio
Bloomberg	**Earnings**	Actual and estimated EPS · Announcement date and time
Bloomberg	**Classification**	GICS sector and industry
yfinance	**Prices & volumes**	630 active tickers — primary tradable universe